

第48届世界技能大赛网络系统管理项目精英班 集训日志（教练）

日期	2026.04.07	教练姓名	黄道金
训练模块	Linux	训练任务	Linux iSCSI server configuration
训练目标	• 掌握在 Debian 13 系统中安装与启动 iSCSI target 服务的方法。 • 理解 iSCSI 中 initiator、target、IQN、LUN、portal 等核心概念及其配置逻辑。 • 掌握使用 targetcli-fb 创建 backstore、发布 target、映射 LUN、配置 ACL 与 CHAP 的方法。 • 能够使用客户端发现与登录测试验证服务端配置，并分析常见访问故障。		
训练内容	• iSCSI server 基础介绍与块存储工作机制说明 • targetcli-fb 安装、服务检查与监听端口验证 • backstore、IQN、LUN、ACL、CHAP 与 target 发布配置 • 客户端连接验证、访问限制与常见问题扩展		
过程记录	Linux iSCSI Server Configuration iSCSI is a block storage protocol that transports SCSI commands over TCP/IP networks. It allows a Linux server to export disks, logical volumes, or image files to remote clients as if they were local block devices. Unlike NFS or Samba, an iSCSI server does not share a directory tree. It publishes storage at the block layer, which means the client can partition, format, and mount the exported device on its own operating system. In this task, we will learn how to build an iSCSI target server on Debian 13, publish a LUN, restrict client access, and verify that an initiator can discover and log in to the exported storage correctly. 1. Task: Install the iSCSI target service and complete the basic deployment. • Install the iSCSI target management tools on the Debian 13 server. • Start and verify the related service components. • Confirm that the server is ready to provide target resources. • Check the default listening port and service status. 1.1. Principles • Which package provides iSCSI target management on Debian 13?		

- Which TCP port is used by iSCSI targets by default?
- What is the difference between an initiator and a target?
- Which commands can be used to verify whether the target service is available?

1.2. Skills

1. Install `targetcli-fb` by using the `apt` package manager.
2. Check service state with `systemctl status target` and inspect the target tree with `targetcli ls`.
3. Verify the listening state of port `3260` by using `ss -ntl`.
4. Confirm that the server has a suitable backend disk, partition, logical volume, or image file before publishing storage.

2. Task: Configure the core features of the iSCSI server.

- Prepare a backend storage object for the target.
- Create an IQN for the iSCSI target.
- Map the backend storage as a LUN.
- Save the running configuration persistently.

2.1. Principles

- What is a backstore and why does the target need one?
- What is an IQN and how should it be named?
- What is a LUN from the client point of view?
- Why should an administrator avoid exporting the system disk directly?

2.2. Skills

1. Create a `fileio` backstore for lab testing or a `block` backstore for a dedicated disk or logical volume.
2. Create a target such as `iqn.2026-04.cn.worldskills:storage.target01`.
3. Map the selected backstore to `tpg1/luns` so that clients can see a usable LUN.
4. Save the configuration to `/etc/rtslib-fb-target/saveconfig.json` so it survives reboot.

3. Task: Configure authentication, permissions, and access behavior.

- Add a portal address for client access.
- Create an ACL entry for a specific initiator IQN.
- Apply one-way CHAP authentication for the authorized client.
- Prevent unauthorized initiators from logging in.

3.1. Principles

- How does an ACL limit which initiator can access a target?
- What is the purpose of CHAP in iSCSI authentication?
- What is the difference between open discovery and restricted login?
- Which mistakes commonly cause login failure on an initiator?

3.2. Skills

1. Create a portal such as `0.0.0.0:3260` or a specific service IP for the target portal group.
2. Disable unrestricted automatic access and create an ACL for the expected initiator IQN.
3. Set CHAP credentials for the ACL entry, for example user `app01` and password `Skills39!`.
4. Verify that only the approved initiator can complete target login successfully.

4. Task: Restrict access, validate security behavior, and complete an advanced scenario.

- Limit target access to a specific client or network path.
- Test discovery and login from an initiator.
- Confirm that a new block device appears only after successful authentication.
- Analyze failure cases by using logs and service inspection commands.

4.1. Principles

- Which checks should be performed when an initiator can discover a target but cannot log in?
- How can logs help identify ACL mismatch, wrong IQN, or wrong CHAP credentials?
- Why is iSCSI more sensitive than file sharing when the same storage is accessed incorrectly by multiple hosts?
- Which advanced topics should be studied after mastering the basic target setup?

4.2. Skills

1. Test the target with `iscsiadm -m discovery -t sendtargets -p <server-ip>` from a client.
2. Use `journalctl -u target` and `targetcli ls` to inspect authentication and mapping problems.
3. Confirm that the client sees a new disk only after a valid login and mapped LUN are present.
4. Extend the design toward LVM-backed storage, multipath, or a dedicated storage VLAN.

	5. Extensions 1. How can LVM logical volumes be used as iSCSI backstores in a training environment? 2. What are the differences between <code>fileio</code> and <code>block</code> backstores? 3. How can <code>nftables</code> be used to restrict access to TCP port <code>3260</code> ? 4. Why is multipath important in enterprise storage networks? 5. How can an iSCSI target be integrated into a cluster or virtualization scenario? 6. References • Wikipedia: iSCSI • Debian Wiki: iSCSI • Debian Manpage: targetcli
总结	训练目标达成情况: 从本次训练完成情况看，选手基本能够独立完成 Debian 13 环境下 iSCSI server 的安装部署，并根据任务要求完成 target 创建、LUN 发布、访问控制和基础验证，说明在网络块存储服务部署方面已具备初步实操能力。 不足之处: • 部分选手对 IQN、ACL、portal 与 LUN 之间的关系理解还不够深入，排查连接失败时容易只关注单一配置点。 • 个别选手能够完成 targetcli 配置，但对 CHAP 认证、后端块设备选择和持久化保存机制掌握仍不够扎实。 改进方向: • 下一阶段应加强基于 discovery 失败、login 失败、无新磁盘出现等现象的反向排查训练。 • 增加 iSCSI 结合 LVM、多路径和高可用场景的迁移练习，提升综合项目应用能力。
填写人	黄道金